

Varsha Hindupur

AI SDE | Software Engineer 3 | Generative AI | Full-Stack

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PROFESSIONAL SUMMARY

Software Engineer 3 at Jack Henry with 8+ years building production systems in regulated financial services. I work across full-stack and platform engineering, and I use Generative AI — RAG, LLM orchestration, and AI-assisted development — in real delivery, including MCP in Cursor to inspect Honeycomb traces without leaving the agent workflow. I help other engineers ramp faster through mentoring, docs, and examples they can reuse. At JH I want to grow as an AI Engineer: shipping prototypes and demos, and staying current by learning as I build.

SKILLS

Languages: Python, Java, TypeScript, JavaScript, C#, SQL, Go, Bash

Web & APIs: React.js, Next.js, Lit, Node.js, Spring Boot, FastAPI, Flask, Django, GraphQL, REST, OAuth/JWT

Data & Cloud: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Snowflake, AWS, Azure, GCP, Docker, Kubernetes, Terraform

AI / ML: RAG, LLMs (Llama, Gemini, Vertex AI), MCP, Hugging Face, ChromaDB / Vector DBs, PyTorch, TensorFlow, AutoML, AI Agents & Agentic Workflows

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (2024) | Microsoft Certified: Azure Fundamentals (2020)

EXPERIENCE

Software Engineer 3 | Jack Henry & Associates, USA

Jul 2025 – Present

- Designed search, verification, member, and account-action screen enhancements as TypeScript/Lit pages on the JH Platform, integrating Picard/Symitar APIs.
- Built Pub/Sub topics and CDC (Change Data Capture) pipelines, provisioning supporting infrastructure with Terraform.
- Engineered a PCI sanitization tool for batch and streaming processing of millions of records, automated through OpCon job scheduling.
- Developed core API functionality, docker container, structured logging, and exception handling for data sanitization tools, reducing execution time from hours to just a few seconds maintaining optimum performance.
- Maintaining and resolving EFM workflow issues by coordinating with the EASE team on correct job configurations.
- Provisioned and set up Honeycomb MCP (Model Context Protocol) server in Cursor Agent, bridging runtime observability to the editor so agent can query Honeycomb and isolate root causes in workflow without switching to UI.
- Continuously building expertise in GCP Vertex AI, Generative AI and Agentic AI development as development enthusiast.

Full Stack Developer | Capital One, Dallas, TX

Feb 2024 – Jul 2025

- Enhanced Rewards Redemption frontend using React and Redux by refactoring component architecture, modularizing UI patterns, and streamlining state management.
- Developed and optimized Java microservices with Spring Boot 3.4.x, integrating JDBC, JMS, JAAS, JAXB, and Hibernate for secure, asynchronous financial workloads.
- Implemented OAuth 2.0 and JWT-based API authentication, reducing unauthorized-access incidents by 20%.
- Used AWS Lambda and API Gateway to support up to 500K API requests per day for Credit Score Analysis and Rewards Redemption with high availability.
- Optimized DynamoDB with indexing and Redis caching for sub-100ms queries; automated CI/CD with Jenkins, GitHub Actions, and Docker.
- Built GraphQL APIs and optimized Python 3.12 Flask/Django services with SQLAlchemy, cutting payload size and improving response times by 300ms.
- Introduced MLOps with Amazon Bedrock, EKS, and MLflow; reduced infrastructure provisioning time by 70% with Terraform and standardized Kubernetes via Helm.
- Delivered domain-specific Generative AI by fine-tuning Hugging Face embeddings and deploying RAG pipelines with ChromaDB for semantic retrieval.
- Orchestrated LLMs (Llama 3.3, DeepSeek R1, Qwen, Gemini) via SageMaker, Bedrock, Groq, and Ollama for low-latency inference in financial applications.
- Mentored junior engineers on Generative AI applications, full-stack delivery, and deployment strategies.

Research, Full Stack Developer — Prof. Kelly Wellman | Northeastern University, Boston, MA

May 2023 – Jan 2024

- Designed a microservices survey platform (MongoDB, Express, React, Node, AWS S3) with an AWS Bedrock Llama 3.3 chatbot, supporting 2M+ users and a 4.7/5 feedback score.

- Engineered auto-scaling on AWS Elastic Beanstalk; led end-to-end full-stack delivery and contributed to a patent effort with stakeholder involvement.

Research, Data Scientist — Dr. Amy Pei | [Northeastern University](#), Boston, MA

Feb 2023 – Apr 2023

- Built a meeting-intelligence app that processed video/audio with PyTorch (92% accuracy) and evolved the model into a multimodal neural net for stronger F1.
- Deployed on Azure App Service with an Azure DevOps pipeline and Blob Storage.

Senior Software Engineer | [Deloitte](#), India

Mar 2019 – Aug 2022

- Developed a credit-risk ML model with XGBoost and TensorFlow at 92% accuracy, reducing false positives.
- Automated 250K+ daily records with Apache Airflow and PySpark, eliminating week-long manual workflows.
- Built a real-time ML prediction system with WebSocket, AWS Lambda, and EventBridge, saving up to \$8,000 per month in infrastructure.
- Led a monolith-to-microservices migration, cutting deployment time from 4 hours to 30 minutes.
- Introduced a TDD automation framework (Pytest, Selenium Grid), reducing defects by 30%; authored technical docs that shortened new-developer onboarding by 2 weeks.

Python Developer | [Capgemini](#), India

Mar 2017 – Mar 2019

- Built a HIPAA-compliant healthcare data system with Python, Pandas, and NumPy, improving workflow efficiency by 40%.
- Launched a Health Information Exchange on Azure Functions and Blob Storage; cut initial frontend load time by 3.2s via lazy loading, caching, and dynamic imports.
- Drove Agile Scrum adoption to accelerate feature delivery.

EDUCATION

Master of Science, Information Systems | Northeastern University, Boston, MA

Bachelor of Engineering, Information Technology | Mumbai University, India

SELECTED PROJECTS

- Patient Triage GenAI Agents — Next.js frontend with Llama 3.2, ChromaDB, and a RAG pipeline to prioritize patients by medical severity.
- E-Commerce on Kubernetes — React (Redux, Material UI) with Terraform, EKS, and ECR; auto-scaling in ~2 minutes; Anthropic-assisted workflows.
- Multi-AZ Cloud Deployment — GCP, Packer, Terraform, Express, PostgreSQL for disaster recovery and high availability.
- Airflow Weather Data Pipeline — Python, Airflow, AWS, PostgreSQL, OpenAI, Redis, and Docker for streaming ETL.